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REGULAR PAPER

Ensemble-based anomaly detection in Brazilian Parliamentary Expenses using unlabeled public data

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Abstract
The rapid growth of digital information offers both opportunities and challenges for monitoring public resource usage. In Brazil, members of the National Congress receive monthly allowances for reimbursable expenses, but manual oversight of this spending is inefficient and error-prone. This work introduces an unsupervised methodology that integrates an ensemble of six machine learning techniques to detect anomalies in parliamentary expenses using unlabeled government data. To guide model training, we leverage six datasets from the ADBench framework, a comprehensive anomaly detection benchmark comprising multiple datasets, selected for their similarity to the target domain. To evaluate the proposed approach, we propose four new real-world unlabeled datasets, which are collectively named the Brazilian Parliamentary Expenses Datasets, obtained from official reimbursement records of the Brazilian Quota for Parliamentary Activity. Results demonstrate that combining multiple anomaly detection methods enhances precision and robustness in identifying irregular expenditures, advancing transparency and accountability in public administration. Moreover, the proposed approach helps prioritize oversight efforts by directing investigations toward cases with a higher probability of irregular behavior.

Keywords Anomaly · Corruption · Machine learning · Government expenditures

1 Introduction

Modern society is increasingly interconnected and immersed in digital data originating from a wide variety of types, formats, and sources. Even simple daily activities, such as paying a bill via Internet Banking, generate substantial volumes of data [1]. Consequently, data generation has grown exponentially, driven by the continuous advancement of new technologies. Devices such as cameras, various sensors, and financial transaction systems are now designed to capture real-world data continuously. This phenomenon is evident when observing the rapid increase in the volume of available digital information over the past decades. Three decades ago, only about 1% of the world's information was available in digital format; today, more than 94% is digitally accessible [2]. These data are produced by a multitude of sources, including mobile phones, servers, sensors, social networks, and the Internet of things.

The proliferation of digital information has, in turn, spurred the development of numerous services and tools that leverage this data, such as recommendation systems and process automation technologies. These services often employ machine learning (ML) techniques to identify patterns, enabling the delivery of personalized experiences, reducing operational costs, and detecting fraudulent activities [1]. The application of ML and monitoring techniques is particularly relevant in addressing corruption, a pervasive problem that affects governments worldwide. Corruption remains a central issue in national and international discourse, with ongoing efforts to curb its spread and ensure accountability [3].

In Brazil, public procurement processes, including contract execution, public tenders, and parliamentary expenditures, are particularly vulnerable to fraud and irregularities. Standard mechanisms enabling corruption include the overpricing of goods and services, the overestimation of

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quantities, and the imposition of excessive qualitative specifications for procured items [3]. Such practices threaten public finances and compromise the quality of life for the population. Thus, ensuring transparency and integrity in public budgeting is essential for fostering a healthy and trustworthy government.

In this context, the enactment of the Brazilian Access to Information Law (*"Lei de Acesso à Informação"*—LAI[1]) in 2011 marked a pivotal advancement in strengthening transparency and fostering closer ties between the government and its citizens [5]. Beyond establishing this legal framework, the Brazilian government has invested in a range of technological initiatives aimed at promoting participatory democracy. These initiatives seek to enhance communication channels between the public administration and the population while improving the quality of public services and products [4]. By encouraging such practices, the government not only advances administrative transparency but also empowers citizens to take a more active role in overseeing public spending and governance. Moreover, the increased availability of data on government expenditures, enabled by the LAI, creates new opportunities for developing innovative methods and tools to monitor and assess how politicians utilize public funds.

In this study, we evaluate various anomaly detection methods and propose a novel approach to investigate the use of public funds by members of the Brazilian Chamber of Deputies. Each member of parliament is entitled to a monthly allowance known as "Quota for the Exercise of Parliamentary Activity" ("Cota para Exercício de Atividade Parlamentar"—CEAP) [6]. CEAP ranges from R$[2] 30,000 to R$ 45,000 and is intended to cover expenses related to their official duties, such as airfare, telephone bills, property rentals, and other parliamentary activities. Given the volume and complexity of the data generated by these expenditures, the development of automated tools capable of detecting anomalies, potential indicators of misuse of public funds, and/or fraudulent practices becomes essential. Detecting misuse often requires identifying patterns beyond simply examining total expenditure amounts. Therefore, more sophisticated anomaly detection (AD) techniques are needed to analyze the combination of features in the data. Another critical challenge is the lack of labeled data indicating past misuse of public funds, which necessitates innovative strategies to evaluate AD methods properly.

Our novel method combines an ensemble of distinct anomaly detection techniques to identify irregularities and potential fraudulent practices based on publicly available expenditure data from members of the Brazilian Parliament. It analyzes different data features and uses an ensemble of AD techniques to identify potential expenditure anomalies with higher precision than traditional anomaly detection methods. For estimating AD performance, we leverage ADBench [7], a comprehensive anomaly detection benchmark comprising multiple datasets, to determine which combination of techniques produces the best results. After analyzing the datasets available in ADBench, we carefully selected six datasets exhibiting characteristics similar to those of our target problem. These selected ADBench datasets were used to train and/or optimize individual AD techniques. Then, we evaluated the performance of each technique and systematically explored all possible combinations of the top-performing ones to identify the most effective ensemble. The final ensemble was then applied to an unlabeled dataset of Brazilian parliamentary expenditures to detect the expenses with a higher risk of being anomalous. By integrating the strengths of different techniques, our approach achieves a more precise identification of irregularities while providing a systematic and reasonable performance estimation process for unsupervised anomaly detection.

The main contributions of this work are threefold:

1. A new set of datasets named Brazilian Parliamentary Expenses Datasets (BPED), constructed from real-world Brazilian CEAP expenditures;
2. The application and evaluation of machine learning AD techniques and statistical methods on BPED, offering a practical framework for identifying suspicious spending patterns; and
3. A novel approach that integrates multiple unsupervised techniques to detect anomalies in parliamentary expenditures, which uses exclusively unlabeled data.

The experimental results indicate that, although more conservative, the technique combination strategy enhances the precision and specificity of anomaly detection. Such high precision is essential in this application domain due to the limited capacity of public institutions to conduct a high number of investigations in parallel.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the proposed methodology. Section 4 discusses the experimental results. Finally, Sect. 5 concludes the paper.

2 Related work

Recent surveys and large-scale empirical studies have reinforced the theoretical foundations of AD techniques, particularly in unsupervised settings where labeled data are rare

[1] The Brazilian Access to Information Law (Law No. 12,527/2011) establishes the right of any individual, without the need to justify, to request and receive information from public bodies at the federal, state, and municipal levels. Its main objectives include promoting transparency, enabling social control over public administration, and fostering accountability [4].

[2] Brazilian Reais (BRL).

or unavailable. The systematic meta-survey by Olteanu et al. [8] shows that the literature lacks a unified taxonomy of AD methods and instead converges on persistent challenges such as unreliable ground truth, sensitivity to modeling choices, high-dimensional effects, and limited benchmarking, highlighting the strongly context-dependent nature of AD. Complementing this perspective, Bouman et al. [9] conducted an extensive benchmark of unsupervised anomaly detection algorithms on real-world datasets and showed that classical methods such as isolation forest, k-nearest neighbors, and density-based approaches remain competitive, but no single method consistently outperforms the others across all datasets [10]. This demonstrates that algorithm performance is dataset-dependent. Therefore, ensemble methods and careful model selection seem to be a promising approach for broadening the range of tasks covered by a single methodology. Similar conclusions are reinforced by recent comparative evaluation studies, which emphasize that effectiveness depends heavily on the data, reinforcing the need for robust evaluation protocols and multi-model approaches when labeled anomalies are unavailable [11].

Several application domains benefit from AD techniques to identify unusual records, such as surveillance [12], industrial settings [13], predictive maintenance [14], rare disease diagnosis [15], financial fraud in the public and private sectors [16], and many others. The approaches for AD, as with most ML techniques, can be categorized as supervised, unsupervised, and semi-supervised. As demonstrated in [7], the performance of AD techniques tends to be superior when labels, even in a small number, are available in the datasets. In most cases, however, there is a vast number of unlabeled datasets that require the use of unsupervised techniques and creative, usually qualitative, methods of evaluation.

Among the different application areas, AD has been extensively explored in the financial domain, particularly for fraud prevention in banking and credit systems. Souza et al. [17] evaluated over 14 techniques (including ensemble models) for detecting credit card fraud. Their experiments, conducted on labeled transactional data, demonstrated that ensemble methods outperformed individual techniques, achieving high precision in fraud identification. While informative, this study relies on fully labeled datasets and focuses on private-sector financial data, differing from the dynamics of public expenditure monitoring.

In the public sector, Niessen et al. [18] applied the isolation forest algorithm to Paraguayan procurement data structured under the Open Contracting Data Standard (OCDS). They assigned anomaly scores to contracts and validated their findings against real-world indicators (e.g., formal complaints), achieving a precision rate of over 90%. Similarly, Sanda et al. [19] investigated Romanian procurement activities and revealed patterns suggestive of collusion, using descriptive data analysis on more than 2 million procedures. These studies demonstrate the potential of AD for oversight in procurement, but they are tailored to contracts and bidding processes, rather than to granular, continuous legislative spending.

Nikola et al. [20] focused on detecting corruption risks in the initial stage of public procurement, specifically during the publication of tenders. Their method involved extracting textual features from tender documents and applying supervised classification algorithms such as Naive Bayes, logistic regression, and support vector machines. The study aimed to identify suspicious "one-bid" tenders by leveraging patterns found in the textual content of bidding notices. While effective within its scope, the approach relies on labeled data and structured procurement documentation, which limits its applicability in settings where records are fragmented, unstructured, or lack outcome labels, such as legislative reimbursement systems.

Ferreira et al. [21] examined parliamentary expenditures by Brazilian deputies between 2015 and 2018. Using unsupervised techniques such as clustering and outlier detection, they uncovered temporal and geographic spending patterns, including peaks at the end of fiscal years and irregularities in transport expenses. While insightful, their evaluation lacks systematic benchmarking or generalizable validation. Moreover, the techniques used are limited in scope compared to recent developments in AD and the study focuses more on pattern discovery than on constructing a replicable framework for AD.

Recent initiatives, such as ADBench [7], address critical limitations in the evaluation of AD algorithms. ADBench compares 30 techniques across 57 datasets, including newly released and real-world corrupted data, using standardized metrics such as AUC-ROC and AUC-PR. Notably, it demonstrates that performance improves substantially with minimal supervision (e.g., 1% labeled anomalies), highlighting the potential of semi-supervised methods. The benchmark also includes tools for ensuring reproducibility and standardizing hyperparameters. Despite these advances, most benchmarking efforts remain general-purpose and do not capture the specific characteristics of public legislative expenditure data, which are marked by free-form justifications, behavioral spending patterns, and a high class imbalance without labeled ground truth.

Overall, the studies reviewed in this section demonstrate the broad applicability of AD methods across various financial, governmental, and procurement contexts. They highlight the potential of these techniques to enhance the integrity of public systems and promote transparency. However, despite these advancements, a gap remains in the literature regarding the specific application of AD to parliamentary spending. Existing approaches often focus on broader public procurement or financial fraud contexts and

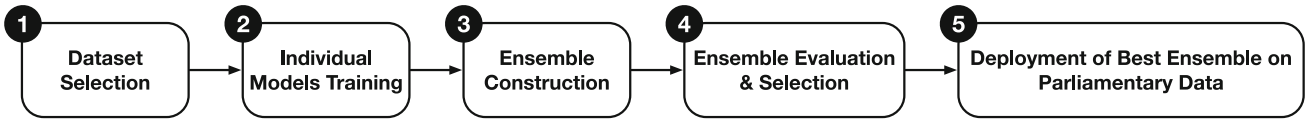


Fig. 1 Overview of the fitting and evaluation workflow for each expense category. For each parliamentary expenditure dataset, a technique is trained and evaluated with ADBench. The top-performing techniques are then recombined to create the final ensemble, which is applied to the unlabeled validation dataset of real expenses

do not provide methods specifically tailored to the characteristics of legislative expenditures.

To address this gap, the present work proposes an ensemble-based AD technique focused on analyzing and detecting anomalies in parliamentary spending datasets. The method is designed to accommodate common features of legislative expenditures and is therefore transferable to parliamentary spending systems in other countries, facilitating comparative oversight and cross-jurisdictional application. Automated tools of this kind are particularly relevant given documented abuses of parliamentary allowances (e.g., the UK MPs' expenses scandal) and the widely discussed role of parliaments in curbing corruption and strengthening accountability. By combining diverse AD techniques and validating them with standardized benchmarks and empirical data, this work aims to provide a complementary instrument for conventional auditing processes and to help oversight bodies prioritize cases for investigation and follow-up [22–24].

3 Proposed approach

This section describes the proposed methodology adopted to create an ensemble-based ML solution for detecting anomalies in parliamentary expenses. We considered a set of widely used unsupervised ML algorithms, chosen for their diverse modeling assumptions and effectiveness in multiple AD tasks. Training and evaluation were performed using the ADBench framework [7], which provides standardized datasets, evaluation metrics, and benchmarking protocols tailored for AD research.

Figure 1 presents an overview of the workflow used in our study. The first step involves training each ML algorithm individually on selected datasets from ADBench. In the second step, we employ ADBench's evaluation procedures to compute relevant performance metrics for each technique. In the final step, the top-performing algorithms are combined into an ensemble, which is then applied to the unlabeled dataset of parliamentary expenses. The subsequent sections detail each phase of this workflow.

3.1 STEP 1: dataset selection and creation

To support both methodological calibration and real-world evaluation, this study relies on two complementary groups of datasets. First, we select and employ a set of preexisting benchmark datasets from ADBench,[3] which provide controlled experimental conditions and reference characteristics commonly used in the AD literature (Sect. 3.1.1). These datasets are used exclusively to guide model selection and ensemble configuration. Second, we introduce a new collection of real-world, unlabeled datasets constructed from reimbursement records of the Brazilian Quota for Parliamentary Activity (CEAP), representing our primary contribution (Sect. 3.1.2). The following sections describe these two groups of datasets.

3.1.1 ADBench datasets

The following six datasets were selected from ADBench: *fraud*, *campaign*, *shuttle*, *cover*, *donors*, and *census*. The selection criteria considered two factors: (i) qualitative similarity to the behavioral and financial patterns observed in parliamentary expenditures, and (ii) variation in anomaly rates and data characteristics that reflect real-world scenarios of class imbalance, sparsity, and feature dimensionality.

Table 1 summarizes the main properties of these datasets. The number of samples ranges from approximately 41,000 to over 600,000 observations, with feature dimensions varying from 9 to 500. The anomaly rates are highly diverse, spanning from 0.17% in the *fraud* dataset to over 11% in the *campaign* dataset. This diversity ensures that the evaluated techniques can generalize across different degrees of sparsity, class imbalance, and feature space dimensionality. Additionally, the selected datasets encompass a range of domains, including finance, sociology, astronautics, and botany, thereby providing a diverse benchmarking environment.

3.1.2 Brazilian Parliamentary Expenses Datasets (BPED)

In this paper, we introduce the **Brazilian Parliamentary Expenses Datasets (BPED)**, a new collection of four real-world, unlabeled datasets constructed from Brazilian CEAP reimbursement records spanning 2019–2023. BPED constitutes a central empirical contribution of this work, providing large-scale, high-dimensional public-spending data that

[3] https://github.com/Minqi824/ADBench

Table 1 Summary statistics of selected ADBench datasets

Dataset	# Samples	# Features	# Anomalies	% Anomaly	Category
Campaign	41,188	62	4,640	11.27%	Finance
Census	299,285	500	18,568	6.20%	Sociology
Cover	286,048	10	2,747	0.96%	Botany
Donors	619,326	10	36,710	5.93%	Sociology
Fraud	284,807	29	492	0.17%	Finance
Shuttle	49,097	9	3,511	7.15%	Astronautics

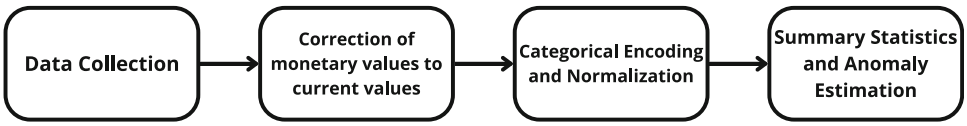


Fig. 2 BPED: step-by-step summary

closely reflect the complexity and noise of real audit scenarios. The datasets cover four major expense categories (food supply, fuel, vehicle rental, and telephony) and range in size from approximately 31,000 to over 281,000 reimbursements per category, with estimated anomaly rates between 4.43 and 8.36% based on IQR analysis. Each dataset combines numerical attributes with high-cardinality categorical variables, such as supplier identity and party affiliation, yielding challenging feature spaces after encoding and strong class imbalance. To support transparency and reproducibility, BPED is publicly available through Kaggle[4] and GitHub,[5] enabling reuse and comparative evaluation by the research community. By releasing and documenting BPED, this work provides a realistic testbed for evaluating unsupervised anomaly detection methods in public-sector auditing, complementing existing benchmarks and enabling reproducible research under conditions that mirror operational fiscal oversight. The construction of the four datasets followed a multi-step data processing pipeline, as illustrated in Fig. 2 and explained in the following paragraphs.

Data Collection

The data were collected using the Jarbas tool [25], developed as part of the "Operação Serenata de Amor" (OSA) project [26]. The OSA initiative seeks to promote social oversight of public administration by analyzing CEAP-related expenses. It leverages data science and engineering techniques, with Jarbas serving as a key component, to provide CEAP records in a simple spreadsheet format, facilitating integration with other applications.

Each dataset comprises a different category of parliamentary expenses:

1. Parliamentary food supply (***ALI***);
2. Fuels and lubricants (***COM***);
3. Vehicle rental or charter (***LOC***); and
4. Telephony (***TEL***).

All datasets share the following attributes (variables): the expense value (BRL), supplier name (from the invoice), party acronym, and invoice date. We also include the supplier's historical average expense value, computed as the mean adjusted reimbursement across all transactions involving that supplier.

Correction of Monetary Values

Since the dataset spans multiple years, all expense values were adjusted for inflation using the Consumer Price Index ("Índice de Preços ao Consumidor"—IPCA) as of February 28, 2024, via the Central Bank's Citizen Calculator.[6] The IPCA is the official measure used to track inflation trends in Brazil.

Categorical Encoding and Data Normalization

To enable ML to process the data, it was necessary to convert the categorical variables into a numerical format. To achieve this, the "one-hot encoding" technique was used for variables such as the supplier's name on the purchase invoice, the party acronym responsible for the expenses, and the date of issuance on the invoice. This approach converts each unique category into a new binary column, where a value of 1 indicates the presence of the category in each row of the dataset, and a value of 0 indicates its absence. This transformation allows the algorithms to adequately capture the relationships between the different categories during training.

Finally, normalization was applied to all variables to place the data on the same scale, improving technique convergence. The technique used was standardization, which arranges all data about its mean and standard deviation of 1. Equation 1 details how normalization was applied:

$$z = \frac{x - \bar{x}}{S} \tag{1}$$

[4] https://www.kaggle.com/datasets/joaopauloschiavon/brazilian-parliamentary-expenses-datasets-bped

[5] https://github.com/SchiavonJP/Brazilian-Parliamentary-Expenses-Datasets-BPED

[6] https://www.bcb.gov.br/meubc/calculadoradocidadao

Table 2 Summary statistics of Brazilian Parliamentary Expenses Datasets (BPED)

Dataset	# Samples	# Features	# Anomalies	% Anomaly	Category
ALI	49,281	5	3,200	6.49%	Food
COM	281,261	5	13,522	4.80%	Car Supply
LOC	31,113	5	1,423	4.43%	Transport
TEL	93,458	5	7,816	8.36%	Telephony

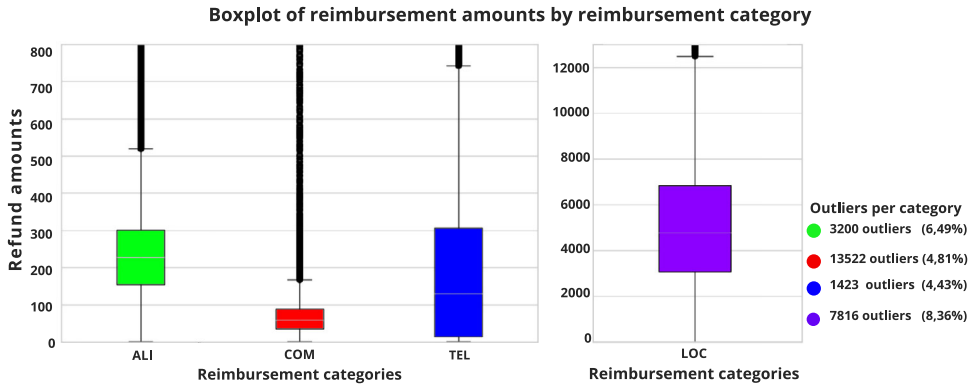


Fig. 3 Boxplot diagram of the reimbursement value data for each category

where z represents the standardized score, x is the value of the observation, $\bar{x}$ is the sample mean, and S is the sample standard deviation.

Summary Statistics and Anomaly Estimation

Since these datasets are unlabeled, the number of anomalies shown in Table 2 was estimated using the interquartile range (IQR) method applied to adjusted reimbursement values. Anomalies are flagged as values that fall outside the range $[Q1 - 1.5 \times IQR,\ Q3 + 1.5 \times IQR]$. Although this method does not provide a ground truth, it offers a reasonable baseline for comparative evaluation and guides the application of unsupervised techniques in real-world contexts.

As summarized in Table 2, these datasets vary in size and anomaly rates, with *TEL* showing the highest proportion of anomalies (8.36%) and *LOC* the lowest (4.43%). The sample sizes range from approximately 31,000 to over 281,000 observations. The COM category shows the highest number of reimbursements, while LOC presents the most significant proportion of duplicate entries (2.5%). Adjusted expenditures in LOC are notably higher and more dispersed, as indicated by the standard deviation and coefficient of variation. The presence of many distinct suppliers in ALI and COM also suggests a wide variation in spending behavior.

Figure 3 shows boxplots of adjusted reimbursement values per category. High outliers are evident across all groups, particularly in ALI and COM. LOC displays broader value dispersion, whereas TEL has numerous high reimbursements that are considered atypical. These anomalies underscore the importance of robust detection methods, as manually inspecting all outliers would be impractical.

Although IQR-based detection helps visualize extreme cases, it tends to produce many false positives by flagging any distant value from the central distribution. Our approach employs an ensemble of ML techniques to reduce false alarms and increase the precision of AD, improving the quality of alerts for fiscal oversight.

3.2 STEP 2: individual models training

After dataset selection, each AD technique was applied to the selected benchmark datasets in a standardized fitting and evaluation workflow. For each dataset, the models were fitted and evaluated using the ADBench evaluation framework, which ensures consistent performance metrics across different methods and data distributions.

Each dataset was partitioned into 70% for training and 30% for testing, using stratified sampling to maintain the original proportion of anomalies. For randomness in the training process, each experiment was repeated three times with different random seeds, and the average results are reported. All evaluation procedures strictly follow the ADBench protocol [7], which ensures consistency and reproducibility through standardized data splits, fixed hyperparameter settings, repeated trials, and well-defined evaluation metrics (AUC-ROC and AUC-PR). The benchmark is fully open-source and provides predefined environments and interactive examples to facilitate replication and extension.

Each selected ADBench dataset was used to independently fit six ML algorithms chosen for their unique assumptions, modeling strategy, and applicability to unsupervised AD tasks. The chosen ML algorithms were selected based on four complementary criteria aligned with the objectives and constraints of this study: (i) suitability for fully unsupervised settings and direct compatibility with score-based ensemble

voting; (ii) high methodological diversity, covering distinct anomaly detection principles; (iii) availability of robust, mature, and widely adopted implementations integrated into the ADBench framework; and (iv) proven practicality and efficiency for large-scale tabular and financial data. Specifically, the selected models represent complementary detection paradigms: distance-based (KNN), density-based (LOF), isolation-based (isolation forest), boundary-based (one-class SVM), probabilistic (Gaussian mixture model), and subspace-based (SOD). This reduces correlated errors and is a standard motivation for ensemble construction. In addition, these methods consistently demonstrate strong performance and computational efficiency in ADBench benchmarks for tabular data, while maintaining interpretability and modest tuning requirements, which are critical for transparency-oriented public-sector applications. Below, we summarize the core principles and characteristics of each ML algorithm considered in this paper:

- **k-Nearest neighbors (KNN)** [27] identifies outliers by measuring the distance of a data point to its k nearest neighbors. The underlying assumption is that normal instances occur in dense neighborhoods, whereas anomalies are distant from their closest peers. Common scoring strategies include using the distance to the k-th nearest neighbor or the average distance to all k neighbors. KNN is nonparametric and straightforward to implement, but its performance is sensitive to the choice of k and suffers in high-dimensional spaces due to the curse of dimensionality;
- **Isolation forest (IF)** [28] is an ensemble-based AD algorithm that isolates observations by randomly selecting features and split values to construct binary trees. Anomalies, being few and different, tend to be isolated closer to the root of the tree. The key insight is that anomalies require fewer splits to isolate compared to normal points. The algorithm constructs multiple isolation trees and computes the average path length for each data point. This average is then used to calculate an anomaly score, where shorter path lengths indicate a higher likelihood of being anomalous. iForest is computationally efficient and particularly suitable for high-dimensional datasets;
- **Local outlier factor (LOF)**, introduced by Breunig et al. [29], is a density-based method that measures the local deviation of a data point concerning its neighbors. Unlike global methods, LOF compares the local density of a point to that of its k nearest neighbors. The local reachability density (LRD) is computed for each point, and the LOF score is derived from the ratio between the LRD of the point and that of its neighbors. A score significantly greater than 1 indicates an outlier. LOF is particularly effective in datasets with varying local densities, providing a more nuanced view of outliers than global distance-based techniques;
- **Gaussian mixture model (GMM)** [30] is a probabilistic model that assumes the data is generated from a mixture of several Gaussian distributions, each with its mean, covariance, and mixing coefficient. The model parameters are estimated using the expectation-maximization (EM) algorithm, which iteratively improves the likelihood of the data under the model. Once trained, GMM assigns each data point a likelihood score; points with low likelihoods are considered anomalies. GMM can capture multimodal data distributions and is effective when the underlying data distribution approximates a mixture of Gaussians. However, it is sensitive to initialization, and the number of components must be carefully chosen;
- **One-class support vector machine (OCSVM)** [31] is a kernel-based algorithm designed for novelty detection in the absence of labeled anomalies. It maps the data into a high-dimensional feature space using a kernel function (typically the radial basis function). Then, it attempts to find the smallest hyperplane that encloses most of the data, separating it from the origin. Data points lying outside this boundary are classified as outliers. OCSVM is effective for scenarios where only "normal" data are available, such as intrusion detection or fraud monitoring, but it requires careful tuning of the kernel and hyperparameters to avoid overfitting or underfitting;
- **Subspace outlier detection (SOD)**, proposed by Kriegel et al. [32], is tailored for high-dimensional data where irrelevant or noisy features can obscure anomalies. The key idea is to identify low-variance subspaces–those dimensions in which a given point's reference set (typically its nearest neighbors) exhibits low variability. An outlier score is computed based on the point's deviation from the mean of its reference set in these relevant subspaces, normalized by the number of such subspaces. This approach mitigates the curse of dimensionality and offers interpretability by highlighting which features contribute to a point's outlier status. SOD is particularly effective in domains where anomalies manifest only in a subset of features.

Table 3 shows the precision and recall metrics of the best individual techniques for each dataset, whereas Table 4 presents the mean precision of each individual technique for all datasets. The obtained results show that GMM achieved the highest mean precision among all individual techniques, with an average of 29%. The fact that GMM alone achieves a mean precision very close to this best-case scenario suggests that it is a strong baseline across multiple datasets, despite the known variability in technique performance.

The results presented in Table 3 are consistent with the findings reported by Han et al. [7] regarding unsupervised

Table 3 Precision and recall scores of the best individual techniques for each dataset

Dataset	Best techniques	Precision (%)	Recall (%)
Campaign	OCSVM	45.8	27.4
Census	KNN	9.3	8.5
Cover	OCSVM	12.4	65.7
Donors	SOD	19.7	39.1
Fraud	KNN	2.2	100.0
Shuttle	OCSVM	100.0	86.1

Table 4 Mean precision scores of individual techniques (six ADBench datasets)

Rank	Technique	Mean precision (%)
1	GMM	29.0
2	OCSVM	28.4
3	IForest	27.6
4	KNN	15.9
5	SOD	13.0
6	LOF	8.5

AD. The authors emphasized that no single unsupervised technique consistently outperforms others across all datasets, underscoring the importance of selecting context-aware techniques. Our findings also illustrate some inherent limitations of unsupervised approaches. As ADBench highlights, a thorough understanding of the underlying data characteristics is crucial for effective AD.

Given this variability and the absence of a clear winner, we propose an ensemble approach utilizing the aforementioned ML models to enhance the precision and robustness of AD. The techniques are grouped into teams of at least three, and their outputs are aggregated. This approach is detailed in the next section.

3.3 STEP 3: ensemble construction

To enhance detection robustness and minimize false alarms, we investigated ensemble strategies that combine predictions from multiple AD techniques. Two voting mechanisms were evaluated:

- **Majority Voting:** A data point is labeled as anomalous if at least half of the techniques in the ensemble classify it as such; and
- **Consensus Voting:** A stricter strategy in which a data point is labeled as anomalous only if *all* ensemble members agree.

All possible combinations involving two or more ML models were evaluated. For each combination, we calculated the mean precision across the six selected ADBench datasets. This systematic comparison allowed us to assess the trade-offs between detection sensitivity and precision, particularly in high-stakes applications such as auditing public expenditures.

Among the tested configurations, the best-performing consensus ensemble achieved higher mean precision than the best majority-voting ensemble. While majority voting offers greater flexibility and typically detects more anomalies, it often increases false positives. The choice to adopt the consensus strategy for the real-world parliamentary datasets was driven by both empirical results and domain-specific needs. In practical auditing scenarios (especially when labels are unavailable), a conservative approach is preferable to reduce noise and avoid overloading human auditors. Consensus voting flags only cases with strong agreement across multiple detection logics, thereby increasing the likelihood, that alerts correspond to genuinely atypical and potentially irregular spending patterns. Consequently, all ensemble results reported in this paper are based on consensus voting.

3.4 STEP 4: ensemble evaluation and selection

Table 5 compares the precision and recall scores of the best ensemble method (using the consensus voting strategy) with those of the best individual technique for each dataset. The results show that the ensemble approach consistently outperforms individual methods, particularly in terms of overall precision and recall. Nonetheless, the relatively low precision observed in certain cases (e.g., 7.1% for the Fraud dataset) aligns with ADBench's findings that unsupervised methods face substantial challenges when dealing with highly imbalanced or complex data distributions.

Table 6 reports the mean precision scores of the top 10 ensemble combinations based on a consensus voting strategy, in which a data point is classified as anomalous only if all techniques in the ensemble agree. The best-performing ensemble (comprising IForest, KNN, LOF, GMM, and OCSVM) achieved a mean precision of 40.3%, underscoring the effectiveness of combining diverse techniques to enhance AD across heterogeneous datasets.

3.5 STEP 5: deployment of best ensemble on parliamentary data

Based on the comparative evaluation of ensemble strategies, we selected the top-ranked consensus ensemble (Table 6), which integrates IForest, KNN, LOF, GMM, and OCSVM. This configuration consistently delivered the highest mean precision across benchmark datasets, confirming its robustness under heterogeneous conditions. We therefore adopted

Table 5 Comparison of precision and recall scores between the best ensemble using the consensus method and individual techniques

Dataset	Best ensemble/single technique	Precision	Recall
Campaign	IF+GMM+SOD	69.7%	6.8%
	OCSVM	45.8%	27.4%
Census	IF+KNN+SOD+OCSVM	12.5%	3.7%
	KNN	9.3%	8.5%
Cover	LOF+GMM+OCSVM	25.0%	34.3%
	OCSVM	12.4%	65.7%
Donors	IF+KNN+LOF+GMM+OCSVM	50.0%	2.2%
	SOD	19.7%	39.1%
Fraud	IF+LOF+GMM+SOD+OCSVM	7.1%	100.0%
	KNN	2.2%	100.0%
Shuttle	IF +GMM+OCSVM	100.0%	78.9%
	OCSVM	100.0%	86.1%

Table 6 Mean precision scores of the top 10 ensemble technique combinations using consensus voting

Rank	Ensemble	Mean precision (%)	Mean recall (%)
1	IForest+KNN+LOF+GMM+OCSVM	40.30	19.60
2	KNN+LOF+GMM+OCSVM	37.80	19.10
3	IForest+KNN+LOF+GMM+SOD+OCSVM	37.20	17.60
4	LOF+GMM+OCSVM	36.30	21.60
5	IForest+KNN+GMM+SOD+OCSVM	36.30	18.80
6	IForest+LOF+GMM+OCSVM	36.20	20.60
7	IForest+KNN+GMM+OCSVM	35.60	23.10
8	IForest+KNN+GMM+SOD	35.30	24.30
9	IForest+GMM+SOD+OCSVM	35.20	22.70
10	IForest+GMM+SOD	34.80	28.10

this ensemble as the final detection model for analyzing the unlabeled Brazilian Parliamentary Expenses datasets introduced in Sect. 3.1.2. The following section presents the outcomes of this application, with a focus on the anomalies identified across different categories of legislative spending.

4 Results and discussion

This section reports the results of applying the proposed consensus ensemble (Sect. 3.5) to Brazilian Parliamentary Expense datasets. We begin by establishing reference baselines derived from national spending statistics, which provide contextual benchmarks for interpreting the anomalies detected by the model (Sect. 4.1). These baselines enable us to relate anomalous reimbursements to typical expenditure ranges and to assess whether observed deviations are economically plausible or suggestive of potential irregularities.

Building on this quantitative context, we then present and analyze the empirical outcomes of anomaly detection across four major categories of legislative spending (Sect. 4.2). Finally, to complement these results and address the lack of large-scale ground-truth labels, we introduce an external validation step based on independent investigations conducted by a civil-society organization (Sect. 4.3). This validation assesses the extent to which model-identified anomalies align with irregularities documented by expert auditors, thereby strengthening the interpretability and real-world relevance of the reported findings.

4.1 Baseline for evaluation: spending patterns

To interpret anomalous reimbursements in economically meaningful terms, we ground our analysis in official statistics and sector-specific benchmarks that reflect typical expenditure patterns across the evaluated categories. According to the Brazilian National Agency for Petroleum (ANP), the average retail price of gasoline in Brazil was R$ 6.11 per liter in 2022. In addition, data from the Institute for Applied Economic Research (IPEA) indicate that the average private vehicle in Brazil travels approximately 9,000 kilometers per year, or about 750 kilometers per month [33, 34]. Based on these estimates, a monthly driving distance of 750 kilometers

corresponds to a fuel consumption of roughly 60–75 liters, depending on vehicle efficiency, which implies an expected monthly fuel expenditure between R$ 370.00 and R$ 460.00. Even allowing for parliamentary travel intensity substantially above the private average, these estimates serve as a reference point for interpreting fuel-related reimbursement values.

For telephony-related expenses, regulatory data from the National Telecommunications Agency (Anatel) indicate an average revenue per user (ARPU) of R$ 49.93 per month for postpaid mobile plans in 2024. This value provides a reasonable proxy for typical monthly reimbursements in the telephony category, particularly for residential and small-business usage profiles [35].

In the case of meals, industry surveys and reports from the Brazilian Association of Worker Benefit Companies (ABBT) estimate the average cost of a standard restaurant meal at R$ 51.61 in 2024. This serves as a practical reference for evaluating per-meal or per-day reimbursement claims.[7]

These empirically grounded benchmarks offer interpretable thresholds and expected value ranges, which we leverage in Sect. 4.2 to contextualize the anomalies detected in the reimbursement data.

4.2 Anomalies detected by the consensus ensemble

This section presents the anomalies detected by the consensus ensemble and compares them against the domain baselines introduced in Sect. 4.1. The goal is twofold: (i) quantify how far flagged reimbursements deviate from typical national spending patterns, and (ii) demonstrate that the ensemble's decisions are driven by multiple features beyond raw monetary value, which results in a considerably reduced and higher-confidence set of cases for manual inspection.

Table 7 and Fig. 4 summarize the monetary magnitude and distribution of anomalous reimbursements identified by the consensus ensemble across the four categories (ALI, COM, LOC, TEL). The ensemble flagged a limited number of cases compared to individual techniques (see Table 8), reflecting its conservative, high-precision design.

When confronted with the domain baselines, several notable deviations are observed. In the meals category (ALI), the ensemble identified 61 anomalous reimbursements with an average value of R$ 2,747.04 and a maximum of R$ 27,691.04. Using the ABBT baseline of R$ 51.61 as a representative per-meal cost, the average anomalous reimbursement is approximately 53 times greater than a typical meal, and the maximum is approximately 536.54 times greater, illustrating large and economically implausible deviations from consumer-level spending.

For the telephony category (TEL), the ensemble flagged 562 anomalies with an average reimbursement of R$ 901.25. Compared with Anatel's reported ARPU for postpaid plans (R$49.93), the average flagged reimbursement is about 18 times the ARPU and about 4 times a conservative business-plan ceiling of R$ 200.00, indicating reimbursements that substantially exceed normal plan costs.

For the vehicle fuel category (COM), the ensemble detected 463 anomalous reimbursements with an average value of R$ 5,199.05. Combining ANP retail price data with IPEA vehicle usage estimates yields an expected monthly private-vehicle fuel expense of approximately R$ 458.25; the ensemble's average anomalous fuel reimbursement is therefore roughly 11 times that expected monthly expenditure.

For the vehicle rental category (LOC), national, directly comparable baselines of routine per-invoice rental costs are not readily available in the same way as for meals, telephony, or fuel; consequently, LOC results must be interpreted with additional contextual information (supplier contracts, region-specific rates, and trip duration) before drawing strong conclusions. The single LOC anomaly flagged by the ensemble (average and maximum R$ 8,946.07) nonetheless stands out in monetary terms and warrants targeted follow-up.

Although the numerical comparisons above emphasize the monetary magnitude of flagged reimbursements, several pieces of evidence indicate that the consensus ensemble does not classify expenses as anomalous solely based on high raw value. First, the ensemble aggregates fundamentally different detection logics (distance-based KNN, isolation-based IForest, density-based LOF, probabilistic GMM, and kernel-based OCSVM). Because the input representation includes one-hot encodings for supplier, party, and invoice date together with standardized numerical features, each technique evaluates distinct aspects of "unusualness." Requiring full consensus forces an observation to be considered anomalous only when multiple complementary criteria concur, which favors cases that are atypical along several independent dimensions (for example, an uncommon supplier combined with an atypical date pattern and an unusual value), rather than those that are merely large in value.

Second, the visual evidence supports this multi-feature behavior. Figure 4 shows that several very large reimbursements are not flagged by the ensemble, while some moderate-sized reimbursements are flagged when they appear in regions of the feature space characterized by sparse local density, uncommon categorical encodings, or temporal clustering. This dispersion demonstrates that the classifier boundaries learned by the ensemble depend on a combination of monetary and non-monetary attributes rather than on a single-value threshold.

Third, the consensus ensemble substantially reduces the volume of cases requiring manual inspection while concentrating on higher-confidence candidates. Using the KNN counts reported in Table 8 as a representative individual-method baseline, the ensemble reduces the set of flagged

[7] https://www.abbt.org.br/home

Table 7 Distribution of anomalous cases identified by the proposed ensemble approach

Category	ALI	COM	LOC	TEL
Average reimbursement value	R$ 2,747.04	R$ 5,199.05	R$ 8,946.07	R$ 901.25
Minimum value	R$ 1.32	R$ 0.65	R$ 8,946.07	R$ 3.34
Maximum value	R$ 27,691.04	R$ 13,712.86	R$ 8,946.07	R$ 6,161.19
Anomalous cases found	61	463	1	562
Percentage of anomalous cases	0.0019%	0.034%	0.0007%	0.071%

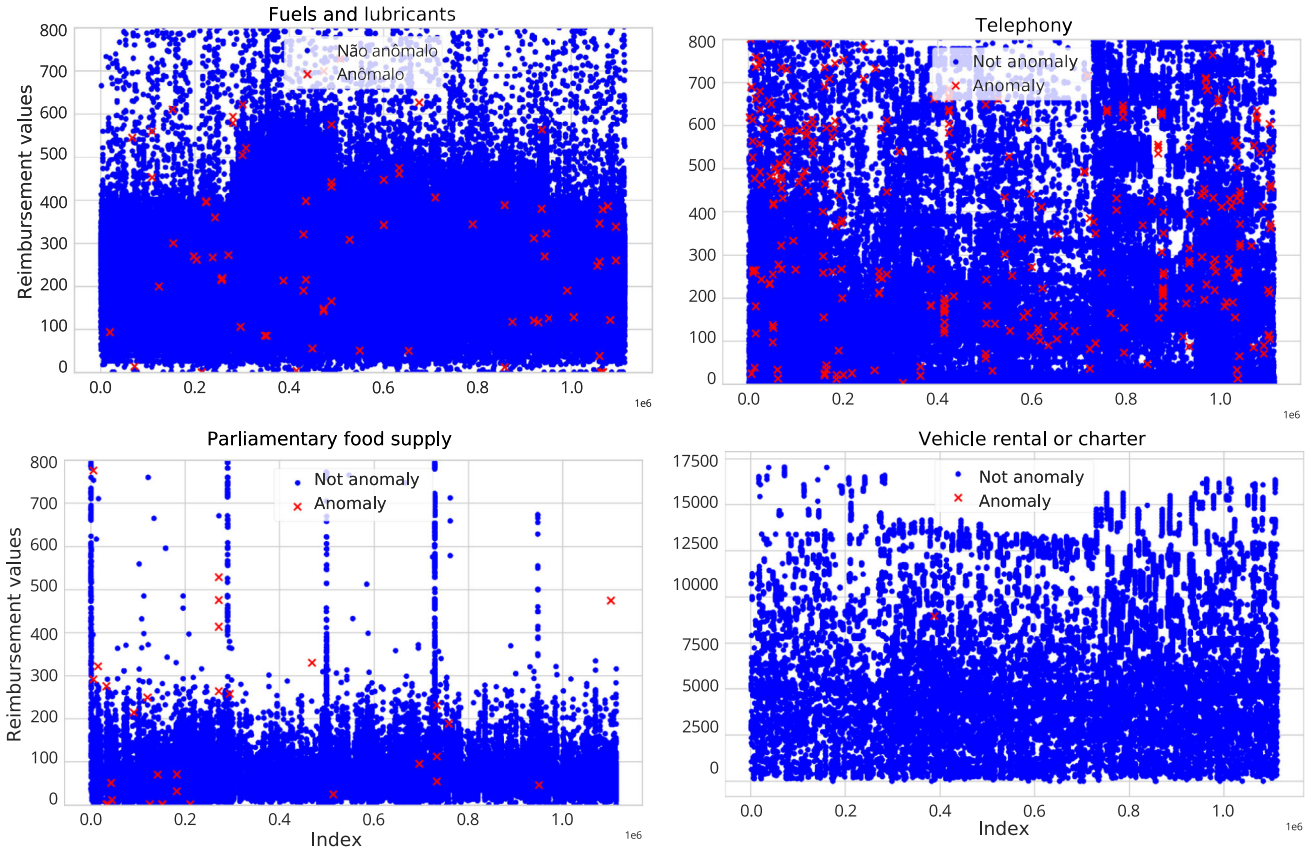


Fig. 4 Scatter plot for each spending category generated from the proposed ensemble approach

Table 8 Number of anomalous cases and percentage relative to the total number of cases by category

Technique	ALI	COM	LOC	TEL
KNN	3,200/6.49%	13,522/4.80%	1,423/4.43%	7,816/8.36%
IFO	3,200/6.49%	13,522/4.80%	1,423/4.43%	7,803/8.34%
MAD	2,960/6.00%	12,406/4.41%	1,204/3.74%	9,380/10.03%
Ensemble	61/0.0019%	463/0.034%	1/0.0007%	562/0.071%

cases dramatically: In the meals category (ALI) flagged cases fall from 3,200 (KNN) to 61 (ensemble), a decrease of about 98%; for fuel category (COM) flagged cases fall from 13,522 to 463, a decrease of about 96%; for vehicle rental category (LOC) flagged cases fall from 1,423 to 1, a decrease of about 99%; and for telephony category (TEL) flagged cases fall from 7,816 to 562, a decrease of about 92%. These reductions illustrate that the consensus ensemble filters out the majority of lower-confidence alerts produced by single techniques, leaving a much smaller and more manageable set of cases for auditors to investigate.

The practical implication of this concentration is that auditors and oversight teams can prioritize limited investigative resources on reimbursements that are simultaneously atypical across several independent signals (value, supplier, invoice-date pattern, local density, etc.), thereby improving

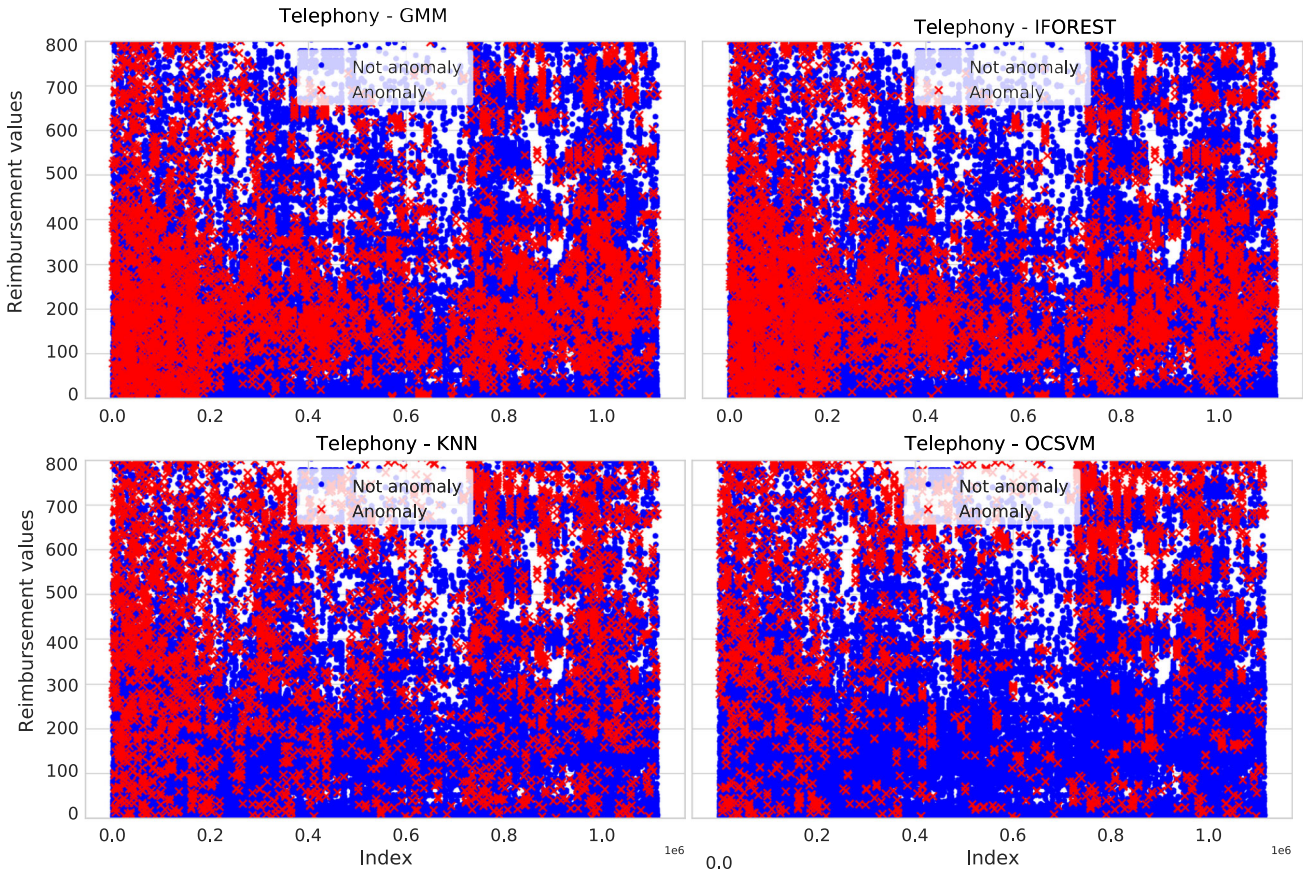


Fig. 5 Visual comparison of individual techniques (example shown for the TEL category)

both the efficiency and the likely yield of manual audits. Figure 5 complements this view by visually comparing the ensemble's output with individual techniques (example shown for the TEL category), highlighting how ensemble consensus filters noise while retaining high-confidence anomalies.

4.3 External validation via NGO investigations

Manually annotating parliamentary expense records at scale is impractical, and reliably determining whether a given expenditure is irregular is inherently challenging. This difficulty stems from heterogeneous documentation, temporal dependencies, and the need for specialized audit expertise. Both suspicious and apparently legitimate expenses often require careful, case-by-case analysis, and sometimes in-depth auditing to support defensible conclusions. Such scrutiny is essential to minimize false positives, as incorrectly flagging legitimate spending can cause reputational harm, distort governance incentives, and undermine fairness, democratic accountability, and institutional credibility. Given the sheer volume of expenditures, this level of analysis makes high-quality, large-scale ground-truth labels exceedingly rare. To address this limitation, we adopt a proxy validation strategy based on independent investigations conducted by *Instituto OPS*,[8] a Brazilian non-profit organization dedicated to public-spending oversight and citizen-led accountability.

Founded in 2018, *Instituto OPS* operates in a strictly non-partisan manner and has received international recognition for its anti-corruption efforts, including the 2020 International Anti-Corruption Excellence Award (Ace Award). Through its auditing initiative, *Operação Política Supervisionada* (OPS), the institute systematically monitors public expenditures at municipal, state, and federal levels by combining technological tools with structured citizen participation. These activities have led to formal complaints filed with oversight bodies and to documented reimbursements of public funds.[9] A central reference for this study is *Operação Tanque Furado*,[10] an investigation targeting fuel-related reimbursements claimed under CEAP by members of the Brazilian Chamber of Deputies. The operation examined 1,863 invoices issued between 2019 and 2020, covering 277

[8] https://institutoops.org.br

[9] https://institutoops.org.br/o-que-ja-fizemos/

[10] https://institutoops.org.br/?s=operaç~ao+tanque+furado

federal deputies. The analysis combined automated screening using the NGO's data-scraping and preprocessing tool (RobOps) with manual audits conducted by trained volunteers.

OPS investigations uncover a wide spectrum of irregularities, ranging from complex behavioral patterns (e.g., implausible usage or unusually frequent daily transactions) to beneficiary-related violations (such as reimbursements involving relatives) and documentation deficiencies. Many of these categories depend on contextual, legal, or external information that is not available in our dataset or represented in the model's feature space. As a result, a comprehensive, one-to-one comparison with all OPS-flagged cases is not feasible.

We therefore adopt a conservative and methodologically sound validation strategy by restricting the analysis to a subset of OPS findings that is fully observable and consistent with our data. This subset comprises cases characterized by unusually high reimbursement values combined with missing or incomplete fiscal documentation. Within this restricted subset, OPS identified 24 parliamentarians with high fuel expenditures and insufficient documentation, all of whom were independently flagged by our ensemble model. This overlap includes at least one documented case in which a parliamentarian reimbursed public funds following an OPS investigation[11] (a case also detected by our model). This confirmed reimbursement provides concrete evidence that the proposed approach can surface irregularities aligned with expert audit outcomes.

To illustrate the nature of the detected anomalies, we present a representative case derived from OPS-documented investigations. In 2020, a federal parliamentarian submitted a high-value fuel reimbursement request totaling approximately R$4,700, supported by a single electronic fiscal document. This document reported the purchase of 1,183 liters of common gasoline in a single transaction. Using the estimations presented in Sect. 4.1, this volume would be sufficient to support an estimated travel distance of approximately 11,800–14,800 kilometers, equivalent to more than 15 months of typical fuel consumption concentrated in a single invoice.

In contrast, normal reimbursement behavior during the same period is characterized by fuel purchases distributed over time, with an average of approximately 4.5 refuelings per month occurring every 1.03 days and multiple invoices of moderate value. On average, the fuel reimbursement amounts to R$ 333.78, which aligns with monthly operational needs and is accompanied by complete fiscal documentation. Even when allowing for parliamentary travel intensity substantially above the private average, the ten deputies with the highest accumulated fuel expenditures over the 2019–2024 period exhibit a comparable temporal pattern, with refuelings occurring on average every 2.5 days and a mean reimbursement value of R$ 499.69, still distributed across numerous transactions rather than concentrated in a single invoice. The ensemble anomaly detection model, therefore, flagged this case due to the extreme concentration of both monetary value and fuel volume in a single reimbursement, independent of any contextual or relational information.

Overall, this proxy validation shows that our ensemble model identifies a subset of high-risk cases that match audit findings. Without an internal ground truth, we cannot estimate real-world precision. Nevertheless, alignment with independently validated OPS cases provides practice-oriented evidence of the model's practical relevance. Future work will involve audit institutions in building manually validated benchmarks and assessing governance, fairness, and reputational risks in real settings.

5 Conclusion

Ensuring the proper use of public funds is a cornerstone of democratic accountability, particularly regarding discretionary expenditures by elected officials, such as members of parliament. However, identifying anomalies or irregularities in such spending remains a considerable challenge, especially in unsupervised settings where ground-truth labels are not available to guide the learning process or validate results. In such contexts, the limited human capacity to conduct thorough investigations and detect misuse of public funds further reinforces the need for methods that minimize false positives while retaining a strong ability to uncover genuine cases of misuse.

In this work, we addressed this challenge by proposing a methodology that combines multiple unsupervised AD techniques into an ensemble model aimed at improving precision and reducing false positives. To calibrate and evaluate the approach, we selected six datasets from the ADBench benchmark that exhibit characteristics analogous to those found in real parliamentary expense records, such as behavioral patterns, class imbalance, and high-dimensional numerical features. The ensemble strategy was then applied to four actual datasets of Brazilian parliamentary reimbursements, covering categories of food supply (ALI), fuel (COM), transport (LOC), and telephony (TEL).

The proposed ensemble combines six techniques: IForest, KNN, LOF, GMM, OCSVM, and SOD. Among the various configurations tested, the consensus-based ensemble formed by IForest, KNN, LOF, GMM, and OCSVM stood out, achieving a mean precision of 40.30%, substantially higher than the best-performing individual technique (GMM, with 29.00%) and also better than the best majority-vote

[11] https://institutoops.org.br/2021/05/28/ops-obriga-mais-um-deputado-a-devolver-dinheiro-aos-cofres-publicos/

ensemble (mean precision of 33.70%). These results confirm that consensus voting is more effective in filtering spurious detections and concentrating alerts on the most promising anomaly candidates.

Visual analyses, including boxplots and density plots, further reinforced the value of the ensemble strategy in reducing noise and prioritizing cases of potential misuse. By synthesizing the strengths of multiple detection methods, the ensemble serves as a filter that highlights only the most suspicious behaviors for closer examination. In general, the results show that the proposed consensus ensemble produces a markedly more conservative set of flagged reimbursements compared to individual techniques, prioritizing high-confidence cases for review. This behavior reduces the number of alerts that require manual inspection and concentrates attention on the most economically implausible reimbursements, which is particularly valuable for resource-constrained oversight teams. Nevertheless, automated detection is only a first step: Human validation, cross-checking with original invoices and supplier registries, and case-by-case contextual analysis remain essential to confirm irregularities and to investigate potential causes.

In summary, this work provides a viable and efficient methodology for unsupervised AD in public-spending data, capable of operating even in the absence of labeled instances. The combination of benchmark-based calibration and real-world application demonstrates the method's generalizability and practical utility in audit scenarios. However, leading with anomaly detection in sensitive public-sector domains raises important governance and ethical considerations that are beyond the empirical scope of this study but must nevertheless be acknowledged. In particular, anomalous behavior identified by unsupervised methods should not be interpreted solely as evidence of wrongdoing or guilt. Rather, the outputs of the proposed approach are better suited to support audit prioritization by highlighting transactions that require closer scrutiny. False positives are intrinsic to anomaly detection, especially in the absence of labeled data, and may arise from legitimate but atypical spending patterns. Our approach aims to mitigate this risk by using a consensus-based ensemble that is deliberately conservative and designed to reduce the likelihood of spurious alerts. However, it does not completely eliminate this risk. In democratic settings, the consequences of false positives can extend beyond individual cases: Premature or uncontextualized disclosure of "flagged" results can erode public trust, distort political competition, and incentivize opportunistic accusations that weaken institutional legitimacy. Even when later scrutiny finds no misconduct, the initial signal can persist in the public sphere, generating durable reputational harm and weakening confidence in democratic representation. For this reason, a strict human-in-the-loop process is essential. Framing anomaly detection as a decision-support tool, rather than an automated judgment mechanism, is crucial to ensure responsible use, preserve fairness, and avoid unintended harm when applying data-driven techniques to public accountability and oversight.

In future work, validating the flagged cases by domain experts or oversight agencies is essential to assess the actual effectiveness of the approach in real-world auditing contexts. Additionally, the construction of labeled datasets and the incorporation of supplementary data sources, such as supplier metadata, time series trends, or text from justification notes, could further enhance AD precision and provide greater interpretability of flagged cases. This would support the development of more reliable and actionable tools to promote transparency and fiscal responsibility in legislative environments. Finally, incorporating more recent deep learning approaches into the ensemble, particularly supervised or semi-supervised methods, is an important direction for applications in which labeled datasets are available.

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Author Contributions J.S. was responsible for implementing all scripts, conducting the experiments, and generating the results presented in the manuscript. M.C. and J.T. contributed to the design of the methodology, the interpretation of results, and the discussion of findings. All authors contributed to the writing of the manuscript, critically revised its content, and approved the final version for submission.

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Data Availability The BPED datasets are available at the following URL: https://github.com/SchiavonJP/Brazilian-Parliamentary-Expenses-Datasets-BPED

Declarations

Conflict of interest The authors declare no conflict of interest.

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